DGMS 2019 • SECTION I: HYDROGEOLOGY & ENVIRONMENTAL GEOLOGY

Question 71: Specific Yield (S_y), Specific Retention (S_r) & Grain Size Controls

Aquifer Storage Fundamentals, Gravity Drainage Physics, and Pore-Size Surface Tension Dynamics

OFFICIAL EXAMINATION QUESTION

Consider the following statements regarding specific yield of an aquifer:

1. It is the ratio of the volume of water that after saturation can be drained by gravity to its own volume.

2. Its value does not depend on the grain size.

Which of the statements given above is/are correct?

(a) 1 only [CORRECT]

(b) 2 only

(c) Both 1 and 2

(d) Neither 1 nor 2

VERIFIED ANSWER KEY

Correct Option: (a) 1 only — Statement 1 is the exact classical thermodynamic definition of **Specific Yield** ($S_y = V_{wd} / V$). Statement 2 is completely false because specific yield depends *profoundly* on grain size: fine-grained sediments (clays) have very small pore throats and huge surface areas where molecular capillary forces retain water ($S_r \approx 48\%$, $S_y \approx 2\%$), whereas coarse sands and gravels release water freely under gravity ($S_y \approx 25\text{--}30\%$).

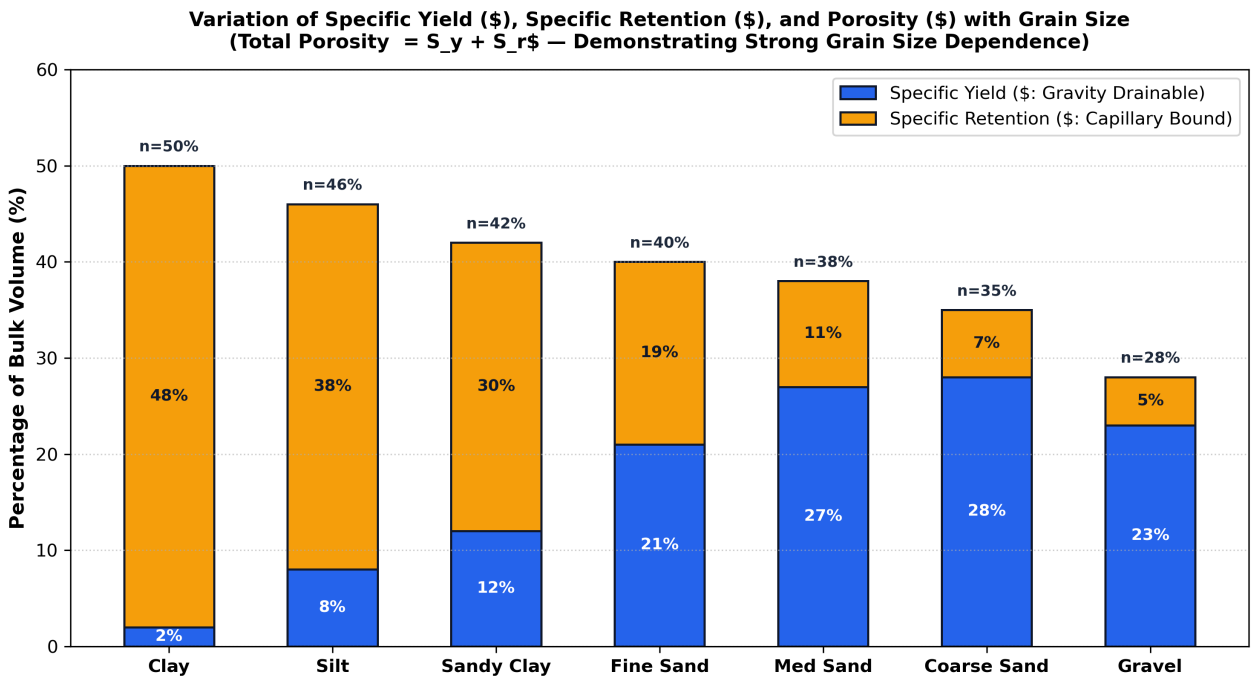


Figure 71.1 | Dependence of Specific Yield (S_y) and Specific Retention (S_r) on Sediment Grain Size: In an unconfined aquifer, total porosity is partitioned into drainable water (S_y) and capillary-retained water (S_r): $n = S_y + S_r$. In clay, high total porosity ($n \approx 50\%$) yields almost no gravity drainage ($S_y \approx 2\%$) because molecular attraction binds pore water. In medium-to-coarse sand, wide pores allow gravity to overcome capillary forces, maximizing specific yield ($S_y \approx 27\text{--}28\%$). (Source: Todd & Mays, *Groundwater Hydrology*; Davis & DeWiest, 1966).

1. Mathematical Definitions in Aquifer Storage

When an unconfined saturated porous medium is subjected to gravity drainage (e.g., during pumping or water table decline), not all water drains out:

- **Specific Yield (S_y):** The ratio of the volume of water drained by gravity (V_{wd}) to the total bulk volume of the rock or soil (V): $$S_y = \frac{V_{wd}}{V}$$
- **Specific Retention (S_r):** The ratio of the volume of water retained against gravity by molecular attraction and surface tension (V_{wr}) to the total volume (V): $$S_r = \frac{V_{wr}}{V}$$
- **Porosity Conservation Law:** The sum of specific yield and specific retention equals the total porosity (n): $$n = S_y + S_r = \frac{V_v}{V}$$

2. Why Specific Yield Depends Strongly on Grain Size

The retention of water in a porous skeleton is governed by **capillary forces** and **adsorption** along mineral grain surfaces.

- The capillary rise height (h_c) in a cylindrical pore is inversely proportional to the pore throat radius (r): $$h_c = \frac{2\gamma \cos \theta}{\rho_w g r}$$ where γ is surface tension and θ is contact angle.
- In **fine-grained sediments** (clay and fine silt), pore radii are microscopic ($r < 2\ \mu\text{m}$), and specific surface area is enormous ($>10\text{ m}^2/\text{g}$). Capillary suction pressures exceed gravity by orders of magnitude, causing virtually all pore water to be held as pellicular and capillary water ($S_r \approx 45\text{--}50\%$). Consequently, S_y drops to an insignificant 1% to 3%, despite clay having the highest total porosity of all sedimentary rocks!
- In **coarse-grained sediments** (medium sand, coarse sand, gravel), pore radii are large ($r > 100\ \mu\text{m}$), specific surface area is low, capillary forces are weak, and gravity drains water freely ($S_y \approx 20\text{--}30\%$).

3. Typical Representative Values in Hydrogeology

Sediment / Rock Type	**Total Porosity (n, %)**	**Specific Retention (S_r, %)**	**Specific Yield (S_y, %)**	**Aquifer Potential**
Clay	45 – 55%	45 – 52%	**1 – 3%**	Aquitard / Aquiclude (holds water, cannot yield it).
Silt	40 – 48%	32 – 40%	**5 – 10%**	Poor yield; low drainage efficiency.
Fine Sand	35 – 42%	15 – 22%	**15 – 23%**	Moderate aquifer yield.
Medium / Coarse Sand	32 – 38%	6 – 12%	**25 – 30%**	**Excellent, prolific unconfined aquifer!**
Gravel	25 – 32%	3 – 8%	**20 – 25%**	Highly transmissive prolific aquifer.

4. Comprehensive Statement Analysis

- **Statement 1: Correct.** "It is the ratio of the volume of water that after saturation can be drained by gravity to its own volume." This is the verbatim classical definition of specific yield.
- **Statement 2: Incorrect.** "Its value does not depend on the grain size." As proved by capillary theory and empirical testing across all sediments, grain size is the primary physical control on the partition between specific yield and specific retention.

5. Authoritative Textbook References

- **David Keith Todd & Larry W. Mays (2005)** *Groundwater Hydrology*, 3rd Ed., John Wiley & Sons, Chapter 2 ("Groundwater Occurrence: Specific Yield and Retention"), pp. 35–45.
- **C. W. Fetter (2001)** *Applied Hydrogeology*, 4th Ed., Prentice Hall, Chapter 4 ("Groundwater Flow: Storage Properties"), pp. 88–95.
- **H. M. Raghunath (2007)** *Ground Water*, 3rd Ed., New Age International Publishers, Chapter 2 ("Hydrogeological Parameters").
- **K. M. Bangar (2018)** *Principles of Engineering Geology*, Standard Publishers, Chapter on Groundwater.